APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Kinematics & Projectile Motion Calculations

SUBJECT: Physics (PHY102)

CLASS & SECTION: Class 11 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 96 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Derive trajectory equation, maximum height, time of flight, and horizontal range with vector calculus integration.

1. Executive Overview & Problem Context

Two-dimensional projectile motion examines the parabolic trajectory of a particle launched into a gravitational field with initial velocity u at angle ;€ relative to the horizontal datum. Assuming negligible atmospheric drag and uniform gravitational acceleration (g = 9.80 m/s² directed downwards), orthogonal horizontal and vertical motions decouple completely.

2. Core Analysis, Solutions & Systematic Working

- **Orthogonal Vector Component Decomposition:**

  Initial velocity u resolves into: Horizontal component u_x = u cos ;€ (constant velocity, a_x = 0); Vertical component u_y = u sin ;€ (uniformly accelerated, a_y = -g). At any elapsed time t: x(t) = (u cos ;,—BÂ ’‡B’ Ò ‡R 6–â ¸)t - 1/2 gt².

- **Derivation of the Parabolic Trajectory Equation:**

  Substituting t = x / (u cos ;,• into the vertical position equation yields: y = (u sin ;,•·€ / (u cos ;,•Ð - 1/2 g[x / (u cos ;,•Û -> y = x tan ;€ - [g / (2u² cos² ;,•Ò ‹"à Since this conforms to the standard second-degree conic form y = Ax - Bx², the physical trajectory is strictly a parabola.

- **Time of Flight (T) & Maximum Height (H):**

  At peak apex, instantaneous vertical velocity v_y = 0 -> u sin ;€ - gt_apex = 0 -> t_apex = (u sin ;,’örà Total Time of Flight T = 2 t_apex = (2u sin ;,’örà Maximum Vertical Elevation H = (u² sin² ;,’ ò ƒ&r’à

- **Horizontal Range (R) & Optimum Launch Angle:**

  Horizontal Range R = u_x × T = (u cos ;,• × [(2u sin ;,’öuÐ = [u² (2 sin ;€ cos ;,•Ð / g = (u² sin 2;,• / g. For a fixed velocity u, range is maximized when sin 2;, Ò ÓâÂ #¸ = 90° -> ;, Ò C[à Furthermore, complementary launch angles (;, æB “ Ò ¸) yield identical horizontal ranges.

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3. Detailed Data Matrix & Systematic Findings

Launch Angle (;,•	Initial Velocity (u)	Time of Flight (T)	Max Height (H)	Horizontal Range (R)
30° Launch	40 m/s	4.08 seconds	20.4 meters	141.3 meters
45° Optimum Launch	40 m/s	5.77 seconds	40.8 meters	163.2 meters (Max)
60° Complementary	40 m/s	7.07 seconds	61.2 meters	141.3 meters (Identical)
15° Shallow Angle	50 m/s	2.64 seconds	8.6 meters	127.5 meters
75° Steep Angle	50 m/s	9.86 seconds	119.0 meters	127.5 meters (Identical)

4. Student Conclusions & Reflective Synthesis

Vector decoupling in Newtonian kinematics demonstrates that independent perpendicular spatial dimensions can be analyzed separately and synthesized via vector addition, underpinning aerospace ballistic trajectory computation.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Problem Solving & Structured Working: 18 / 20

Analytical Depth & Research Methodology: 28 / 30

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding kinematic mastery! Mathematical derivations, vector decomposition, and range proofs are flawless."

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